A browser-based seismic observatory for regional earthquake catalogue comparison in North America

José Manuel Villa Vargas; Gabriel Hurtado Avilés[1]*

[1] Escuela Superior de Cómputo (ESCOM), Instituto Politécnico Nacional, Mexico City, Mexico

*Corresponding author: Gabriel Hurtado Avilés. Email: ghurtadoa@ipn.mx.

ORCID: José Manuel Villa Vargas, https://orcid.org/0009-0009-2401-0661; Gabriel Hurtado Avilés, https://orcid.org/0009-0002-5686-1822.

Abstract

Earthquake catalogues differ in coverage, measurements and revision histories, complicating regional comparison and reuse. We present a browser-based seismic observatory connecting a computational system, documented inputs and reproducible descriptive results. Four frozen USGS, NRCan and SSN catalogues supply 287,936 source records. Explicit adapters preserve reported magnitude types, missing values and provenance without reconciling physical events. Verified static delivery supports maps, administrative selection, statistics, historical rankings, spatial-pair queries and exports without Docker or a database server. A 659-record USGS example returns mean magnitude 5.311; matched Canadian and Mexican land selections contain 15 and 53 records. A separate 36-event ComCat pilot identifies magnitude changes in 13 of 31 comparable retained endpoint pairs and four crossings of 5.5. Independent SQL agrees with 4,720 conditional retained projections. Executable evidence contracts separate captured observations from archive-dependent calculations and unsupported historical reconstructions. Results remain descriptive: completeness, source accuracy, physical independence and usability are not established, and no earthquake forecast or seismic-risk estimate is produced.

Keywords: earthquake catalogues; geovisualization; provenance; retained revisions; reproducibility; evidence contracts

1. Introduction

Regional earthquake analysis must address heterogeneous measurements, duplicates and magnitude compatibility (Weatherill et al., 2016); Mexican hazard studies additionally require completeness assessment and declustering (Sawires et al., 2019). Displaying several catalogues together does not perform those operations.

The problem is to turn incompatible regional earthquake records into inspectable selections and reusable descriptive information. We connect a browser-based system, documented seismic inputs and traceable results for study and further research. The inputs are frozen snapshots, not live monitoring.

ComCat summaries and versioned products evolve (U.S. Geological Survey, n.d.a-c; Thompson et al., 2020). Libcomcat, Timeline and INGV offer history capabilities (U.S. Geological Survey, n.d.d,e; Pintore, 2026). Our evidence contracts follow resolution-aware query research (Sismanis et al., 2009; Gardezi and Bertossi, 2011), bounding calculations when historical preferences/completeness are unknown. Documented inputs extend Villa Vargas et al. (2026); Figure 1 connects the workflow.

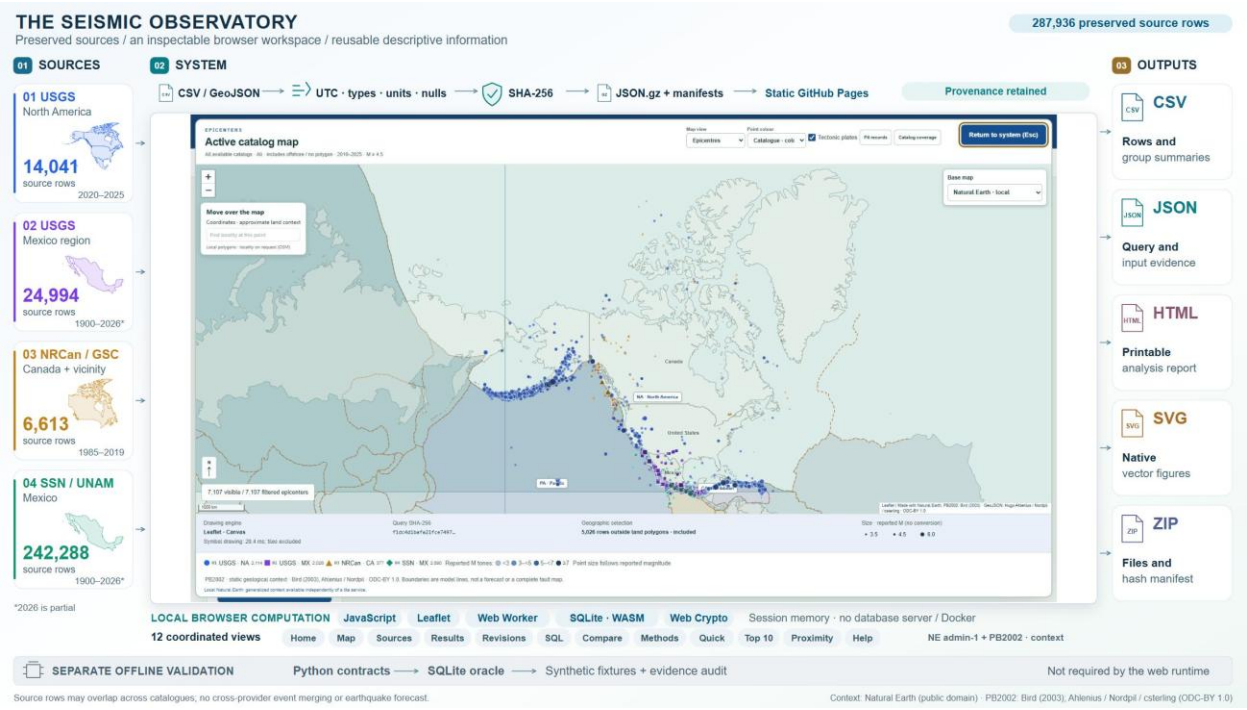


Figure 1. Implemented architecture connects four source packages, explicit adapters and verified static delivery to browser maps, analysis and reusable exports. The central map shows 7,107 selected rows (2010-2025, $M \geq 4.5$, all catalogues); inputs total 287,936. Optional SQLite/WebAssembly runs in a Worker; country/subdivision indexes and PB2002 supply separate context. The offline Python/SQLite audit is distinct from the browser runtime.

Interactive view: Open implemented design.

2. Computational system and analytical methods

2.1. Static architecture and reproducible delivery

Python adapters package preserved CSV inputs as JSON with SHA-256 manifests. GitHub Pages serves HTML, CSS, JavaScript, Leaflet 1.9.4 and bundled dependencies. Verified packages occupy browser session memory; no Docker, database server or persistent browser database is required. Home initially displays all 287,936 source rows (Figure 2). English is the default, with Spanish, themes and light/dark modes available.

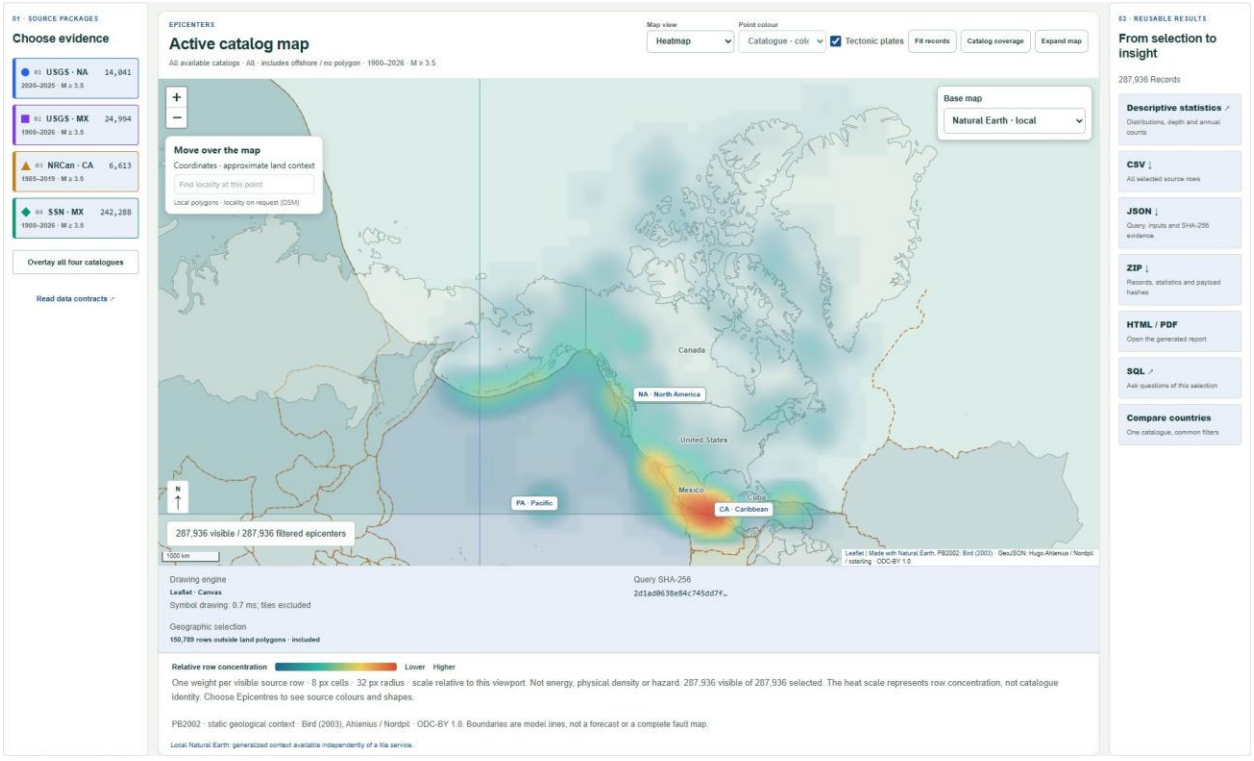


Figure 2. Home initially selects all 287,936 rows across four catalogues. Point and optional heat views retain the same query and exports. Heat displays relative screen-space row concentration, without magnitude weighting or hazard estimation. PB2002 adds static tectonic context; spatial coincidence does not establish a mechanism.

Interactive view: Open this view.

Gzip companions bind compressed and original lengths/hashes, verified before parsing. Loading includes progress, bounded retries and cooperative processing. Unsupported gzip or HTTP 404/415 permits verified JSON fallback; integrity failures stop loading. Session caching avoids repeated work.

2.2. Normalization and geographic selection

Adapters align identifiers, labelled UTC times, coordinates, reported magnitudes and documented depths (Figure 3). They retain raw fields, case, empty strings and unknowns; absent common fields become null. No Mw conversion occurs. Undocumented Canadian depths stay outside kilometre statistics; SSN keeps UTC/local text. ComCat IDs survive; SSN/NRCan receive content-hash row keys with occurrence suffixes. Atlas and Time and Coverage share source profiles. Hue/shape identifies catalogue; lightness bands encode reported M < 3, 3-<5, 5-<7 and ≥7, without scale equivalence.

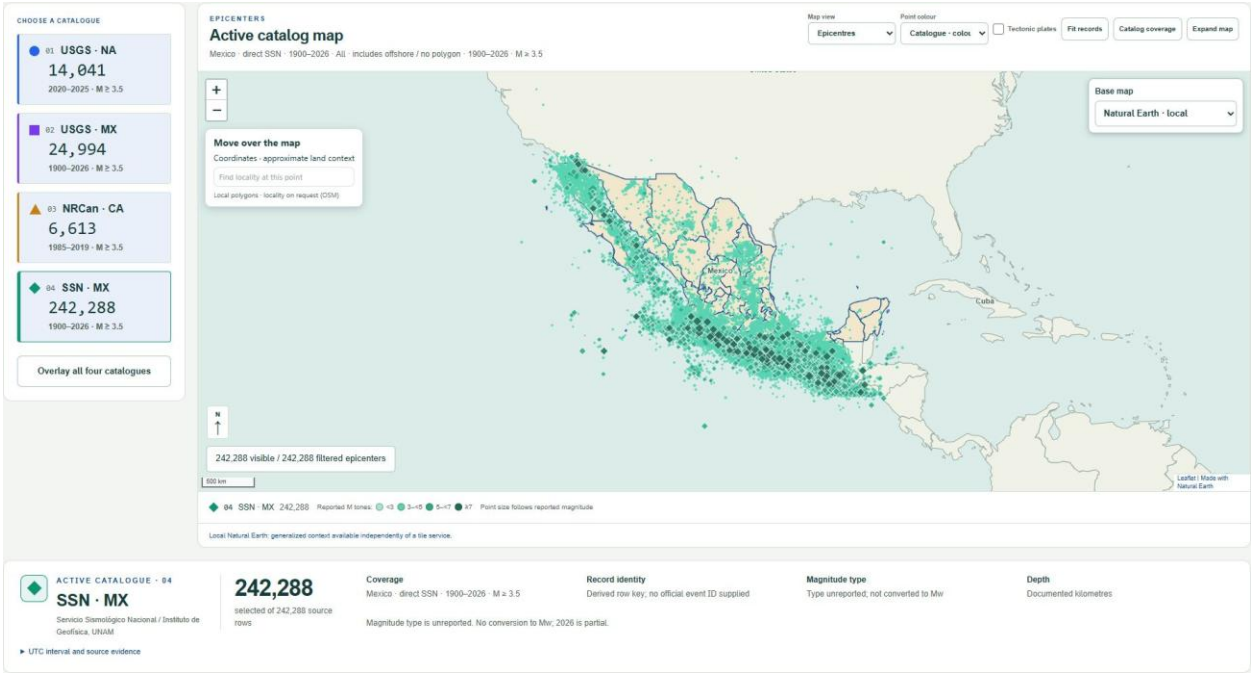


Figure 3. Sources Atlas displays the SSN package (242,288 rows), a regional map and a four-source rail. The complementary Time and Coverage view is accessible through its button. Hue/shape identifies source; darker tones indicate larger reported magnitudes within that hue. Four packages from three distributors are not independent observing agencies. Interactive views: Atlas · Time and coverage.

Composition retains registry order and catalogue-qualified keys. It flags 2,163 shared ComCat IDs without merging rows or matching agencies' events by proximity. Thus 287,936 rows are not asserted independent earthquakes. Natural Earth provides 47 generalized 1:50-million land classes and 743 pinned 1:10-million administrative units (Natural Earth, n.d.a,b). Mexico has 32 named units and a minor-island unit. Indexes bind geometry, event hashes and row order. Confirmed country clicks enable subdivision refinement. Offshore, boundary and unavailable classes

stay separate; land containment implies neither jurisdiction nor nationality. Mexican state context also displays without excluding offshore rows (Figure 4).

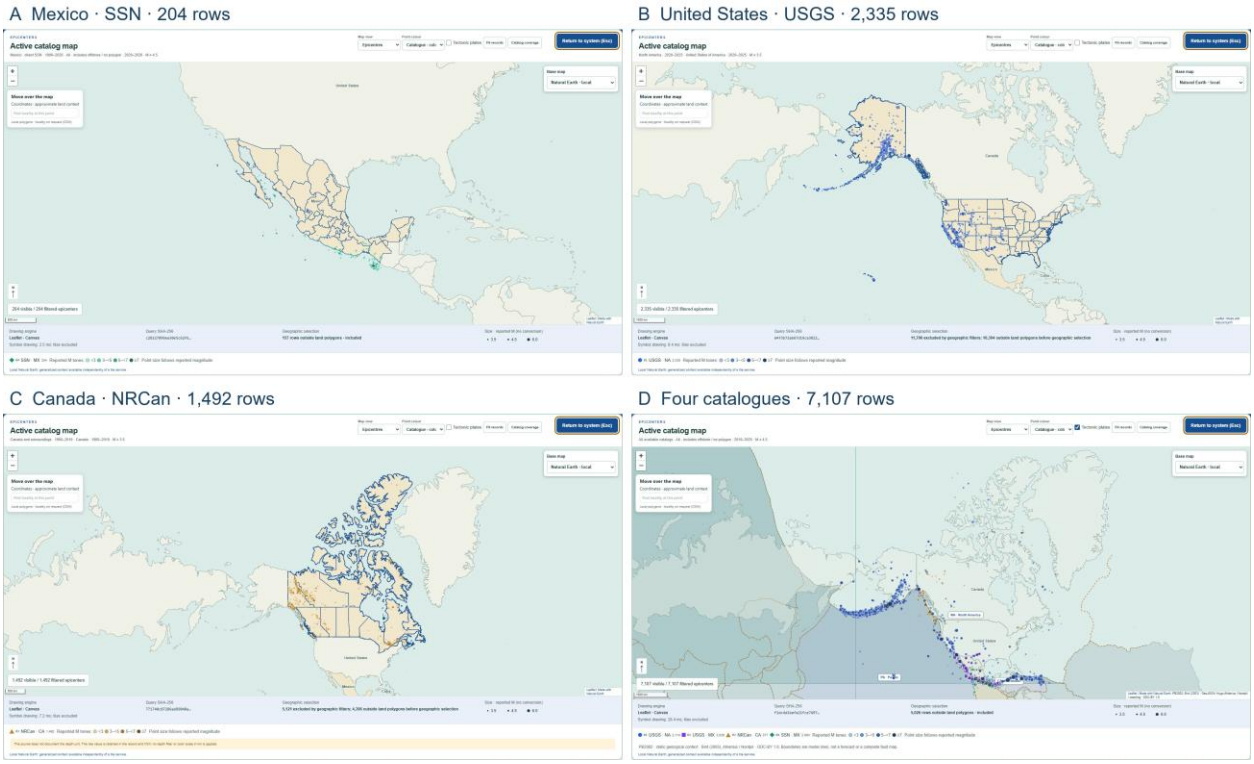


Figure 4. Four distinct map selections. (A) SSN, partial 2026, M ≥ 4.5, all classes: 204 rows, with Mexican subdivision context but no land filter. (B) USGS North America, 2020-2025, M ≥ 3.5, USA land: 2,335. (C) NRCan, 1985-2019, M ≥ 3.5, Canada land: 1,492. (D) All catalogues, 2010-2025, M ≥ 4.5: 7,107. Counts are source rows; map extent does not restrict exports. Interactive views: A · B · C · D.

Regional queries request UTC 2000-2026 and M ≥ 3.5 (Figure 5): Guerrero/SSN has 19,963 rows; California/USGS, 551; Yukon/NRCan, 225; Alaska/all catalogues, 1,112. Available coverage is 2000-partial 2026, 2020-2025, 2000-2019 and 2000-2025 respectively. Empty uncovered years are not evidence of seismic inactivity.

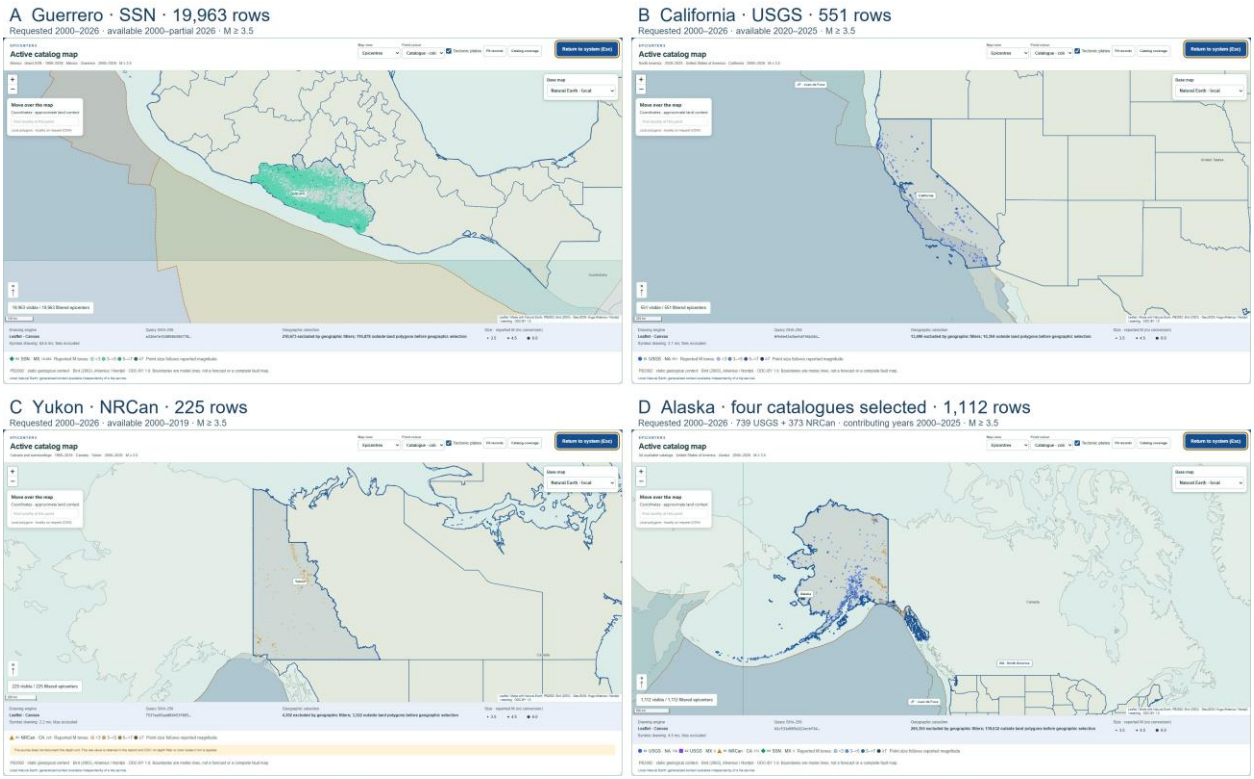


Figure 5. Subdivision queries, requested UTC 2000-2026, reported M ≥ 3.5 and land containment: (A) Guerrero, SSN, 19,963 rows (2000-partial 2026); (B) California, USGS North America, 551 (2020-2025); (C) Yukon, NRCan, 225 (2000-2019); (D) Alaska, four catalogues selected, 1,112 (739 USGS + 373 NRCan; 2000-2025). Archive limits remain visible; these are not complete 27-year national histories. Interactive views: A · B · C · D.

Filters cover UTC years, reported magnitude/type, agency, documented depth, missing horizontal error and ID/place text. Coordinated views follow Wu et al. (2020). Local Natural Earth, OSM, humanitarian OSM, USGS Topo and static August-2004 NASA Blue Marble backgrounds are selectable on each map, including historical tools. Remote tiles load on demand with attribution. Viewport and display-longitude shifts never alter exported coordinates or selection. The query digest binds inputs, filters, geography and ordered keys; it is not a digital signature.

Heat mode assigns one weight per visible source row to 8-pixel CSS cells, then smooths with a 32-pixel Gaussian radius and sigma 12 pixels. Log1p colours are relative to the viewport maximum, with clipped edge kernels. Zoom changes displayed concentration, not physical density, energy or hazard. Point mode preserves the same selection. The coordinate panel reports local approximate land context; an explicit cached OSM lookup supplies an approximate locality, not a surveyed address. For linked USGS M ≥ 6 records, official ShakeMap and DYFI links distinguish estimated intensity from felt reports. No propagation or felt radius is inferred from magnitude.

2.3. Descriptive tools and historical queries

Statistics use one catalogue and exact reported magnitude type; unknown types form a separate numerical group without asserted homogeneity. Means, medians, ranges and sample standard deviations retain their valid counts. Depth requires documented kilometres. Pearson correlation requires a known type, three finite pairs and nonconstant variables; unavailable values remain null. Annual zero bins mean no selected rows, not physical absence.

Country comparison applies common UTC years, threshold and type within USGS North America 2020-2025. Histograms share lower-inclusive/upper-exclusive 0.25-M bins; annual points retain discrete counts. No area/population normalization or equivalence of SSN/NRCan coverage is assumed.

Top 10 automatically queries all four catalogues; country/state controls refine the scope. It ranks heterogeneous reported magnitudes, not homogeneous Mw or the strongest-ever population. Shared exact ComCat IDs yield one entry with both provenances. Before filtering, the later valid UTC update chooses the representative; invalid/tied times prefer North America then Mexico, never the larger magnitude. SSN/NRCan are not matched across agencies.

Advanced controls restrict catalogue/type/period. Bounded selection retains ten entries, ordered by descending M, ascending UTC time and key; additional ties are omitted. Offset rank labels use leader lines to the original epicentres, leaving coordinates unchanged.

Historical proximity enumerates distinct-key pairs within one catalogue/type/UTC interval, default $M \geq 7$. Haversine uses $R = 6{,}371.0088$ km and 25/50/100/200-km radii. Optional 100-km offshore context uses minimum spherical point-to-minor-arc distance to generalized land boundaries without annexing assigned neighbours. Both predicates allow 0.000000001 km numerical tolerance before rounding, not geographic uncertainty. Direct pairs imply neither transitive groups, repeated ruptures, recurrence intervals nor risk.

Optional PB2002 supplies static plate/boundary context (Bird, 2003; tectonicplates contributors, n.d.), without inferring mechanisms or active faults. Guided sql.js 1.13.0 SQLite/WebAssembly queries run in an isolated Worker (sql.js contributors, n.d.). Temporary relations preserve qualified keys and null identifiers/depths; query-only execution permits one SELECT/WITH statement, cancellation, timeout and 5,000 returned rows.

Simple view gives three direct routes to maps, statistics and reports using the full engine. Help covers sources, version, contact and licensing. CSV, JSON, HTML and SVG retain scope; ZIP adds payload hashes. Printable HTML supports PDF. Query schema seismic-local-query/5 preserves geographic evidence and supported older URLs. Functional checks do not constitute a usability study.

2.4. Contracts for retained revisions

The separate offline audit identifies lineages by (source, type, code) and versions by updateTime. Exact four-part repeats are deduplicated; conflicting content fails. Distributor, origin author and magnitude author remain distinct. Primary endpoints require source us, type origin and code equal to the captured event ID; internal-origin and phase-data are excluded.

Table 1 bounds three output contracts. Projection selects each lineage's latest retained producer time at/before a cutoff, reports DELETE separately without reviving predecessors, then filters attributes. Filtering first can preserve obsolete above-threshold versions. Nulls fail value predicates; types retain case, without magnitude conversion.

Table 1. Contract	**Interpretation of the output**	**Principal boundary**
`snapshot_observed`	Query result on supplied current summary bytes at capture	Not a historical replay or a complete physical population
`retained_projection_conditional`	Latest retained version per lineage at or before a producer-time cutoff	Neither public availability nor historical preference is established
`historical_catalog_not_identifiable`	Stronger historical catalog claim is declined	Missing historical evidence is not manufactured

Unknown membership, release times, preferences and associations prevent complete historical public-state reconstruction. Users acknowledge retained-only evidence and distinct producer/publication clocks. Declining a stronger claim does not assert that a particular missing version existed.

Endpoints are earliest/latest retained non-deletion products, not first release or truth. Magnitude subtraction requires identical literal types; inclusive thresholds are 5.5, 6.0, 6.5, 7.0, 7.5 and 8.0. Haversine measures separation, not accuracy gain. Adjacent versions compare 27 declared parameter/agency/uncertainty fields; unchanged fields do not imply identical products.

2.5. Independent evaluation

SQLite independently normalizes original products and ranks before filtering, using Decimal magnitudes and integer UTC times. Ten specifications test thresholds, literal mww/mb/mB, bounds, earthquake type and half-open 2017/2022 intervals immediately before/at retained updates. Input hashes and lengths are checked.

Three synthetic paired histories have identical current evidence but different past release, preference or withdrawn membership. Eight further histories test missing revisions, deletion and clock/filter-order substitutions; regional/magnitude variants yield 24 cases. Independent public-action replay supplies synthetic truth. These tests establish logical ambiguity, not loss prevalence. Comparator documentation supplies context, without speed/usability benchmarking; the maintained Timeline destination was inaccessible.

3. Seismic inputs and study design

Four catalogue packages from three distributors demonstrate exploration separately from revision validation (Table 2). Their coverage is not a sequence of historical public states.

Table 2. Catalogue and distributor	Origin-time coverage	Records	Interpretation limits
Mexico-region diagnostic / USGS ComCat	1900-2026 (partial)	24,994	Earthquake records; regional rectangle; two quarry blasts excluded; not an independent SSN catalogue
North American regional windows / USGS ComCat	2020-2025	14,041	Earthquake records from twelve pinned CSVs; regional windows, not national assignments or a revision archive
Canada and surroundings / NRCan/GSC	1985-2019	6,613	Reported M ≥ 3.5; five repeats retained; original event IDs, event types and uncertainty/revision fields absent
Mexico and surroundings / direct SSN	1900-2026 (partial)	242,288	Reported M ≥ 3.5 selected from 402,082 source rows; magnitude types, original IDs and revision fields absent; coverage is not completeness

Both Mexican packages include partial 2026. USGS North America covers regional/Aleutian windows (14-84° latitude, -180 to -50° longitude; Aleutians 50-73°, 170-180°). Original positive longitudes survive for 770 rows. The 6 November 2026 Canadian download retains 6,613 of 101,365 rows at M ≥ 3.5, including five repeats. IDs, event types and depth units remain unknown. This resource carries the Open Government Licence-Canada, not automatically the NEDB bulletin's terms (Natural Resources Canada, n.d.; Government of Canada, n.d.). The SSN download covers 1900-2026-11-06, all magnitudes/depths/states (Servicio Sismológico Nacional, 2026a). Ten fields retain UTC/local times, magnitude, coordinates, kilometre depth, locality and status. Among 402,082 rows, 384,433 have numeric magnitudes; 17,649 are uncalculable; 242,288 satisfy M ≥ 3.5. IDs, magnitude/event types, uncertainty and revisions are absent. Verificado/revisado are statuses, not histories.

The source generation label implies 19:57:07 UTC, preceding the latest reported origin, 20:31:07 UTC. Eight UTC/local pairs have atypical offsets; neither column is corrected and the label is not an atomic capture. Gaps and tied annual maxima remain. SSN credit, meaning and reproduction terms are retained (Servicio Sismológico Nacional, 2026b). The documented 1988 reporting change and M_d 4.3 completeness for 1988-1998 (Zúñiga et al., 2000) do not establish completeness here.

The pinned upstream snapshot 8fe055ffe635875f7290906d45f70562a7896621 has 24,996 IDs across 46 CSVs; two quarry blasts are excluded. UNM/us_unm attribution selects 9,858 earthquakes. UNM identifies SSN/UNAM authorship, while USGS distributes the records (European-Mediterranean Seismological Centre, n.d.).

The separate ComCat pilot selects current 2017/2022 earthquakes with preferred $M \geq 5.5$, latitude 14-33° and longitude -118 to -86°. Retrieval on 5 November 2026 spans 19:53:15.316-19:53:54.609 UTC. Checked count/detail responses (format=geojson, includesuperseded=true) retain bytes, URLs and hashes. Current selection omits formerly qualifying events outside today's criteria, precluding estimates of all historical withdrawals/crossings.

4. Results from reproducible selections

4.1. Regional information and reuse

Source profiles expose package coverage, fields, verification and downloads. Distinct map selections return 204 partial-2026 SSN rows at $M \geq 4.5$, 2,335 USGS USA-land rows, 1,492 NRCan Canada-land rows, and 7,107 combined rows under their stated filters (Figure 4). Distributor choice and geographic containment remain separate.

Figure 6 shows a guided SSN Guerrero land selection, UTC 2020-2025, $M \geq 3.5$. Its 7,930 rows yield mean reported M 3.691, median 3.6 and mean documented depth 31.17 km. Summary measures and six annual bins share that selection; absent magnitude types do not establish a homogeneous scale.

The independent benchmark selects 659 USGS rows, UTC 2020-2025, $M \geq 4.5$, literal mww, all geographic classes: mean M 5.311168, median 5.2, range 4.5-8.2, sample SD 0.523513. Documented depths have mean 32.832339 km and median 15.14 km; Pearson r = -0.003260 implies neither independence nor causality. Python matches JavaScript and ordered IDs within 10^{-12}; exports retain criteria/hashes.



Figure 6. Guided descriptive results for 7,930 SSN Guerrero land records, UTC 2020-2025, M ≥ 3.5. Summary cards, the 0.2-M histogram and six annual bins use the same selection. Reported magnitude types are absent; numerical grouping establishes neither scale homogeneity, completeness nor physical independence. No occurrence-rate or risk estimate is presented. Interactive view: Open these results.

Administrative selection changes the query; plate display does not. The enlarged UTC 2000-2026 view contains 203 SSN Guerrero land rows at M ≥ 4.5 (partial 2026; Figure 7). Neither supplies location uncertainty or mechanisms.

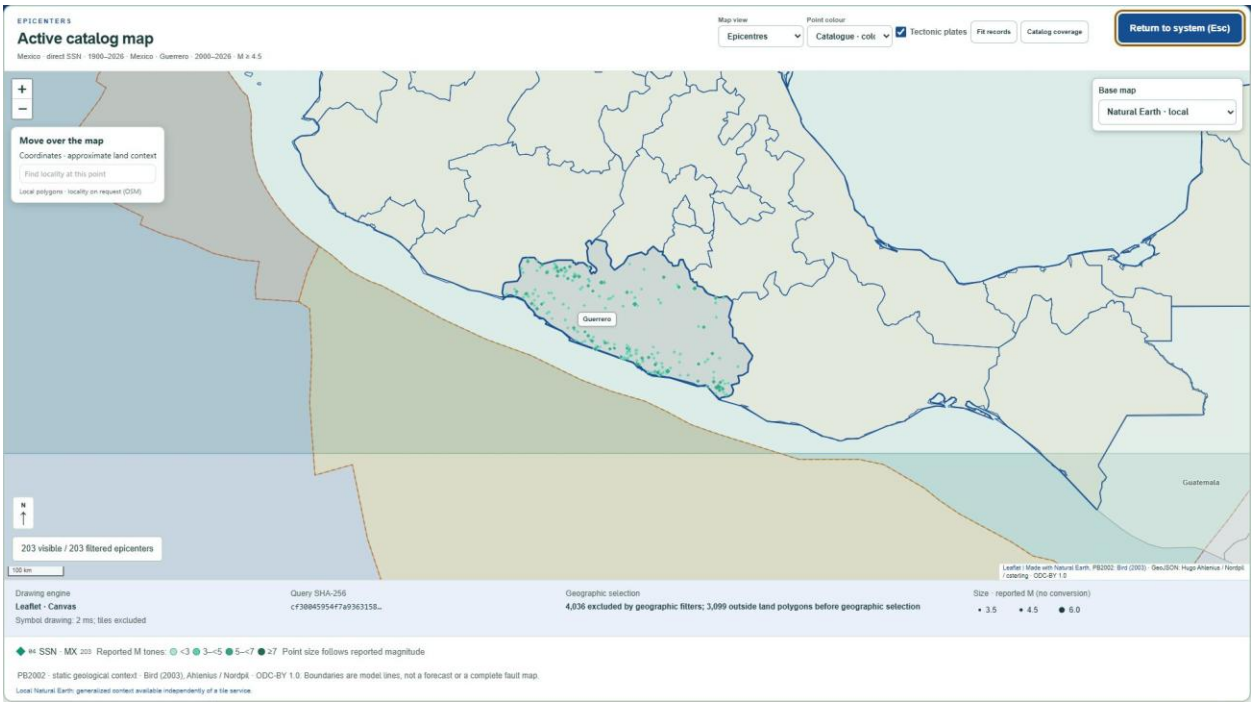


Figure 7. The enlarged Guerrero map shows 203 SSN land rows, UTC 2000-2026 (partial), M ≥ 4.5, administrative limits and static PB2002 context. Confirmed country/subdivision filters alter the query; plate display does not reclassify records or supply mechanisms. Natural Earth classes are generalized land containment, with unassigned/offshore rows retained separately. Interactive view: Open this zoomed map.

Under the same USGS period, threshold and mww type, Canadian and Mexican land classes contain 15 and 53 rows (Figure 8). Shared bins and annual counts describe selected records; unequal areas, offshore exclusions and unassessed completeness preclude relative-hazard or occurrence-rate comparisons.

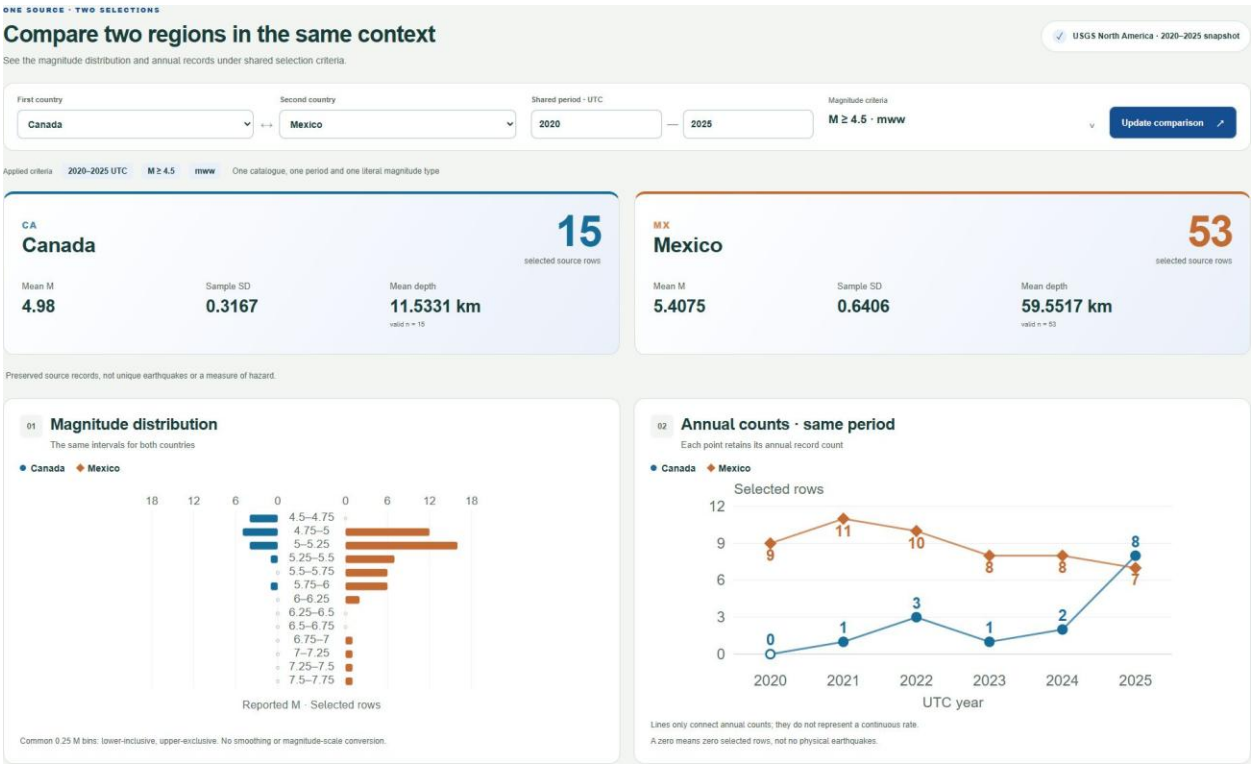


Figure 8. Canadian and Mexican land selections within USGS North America, UTC 2020-2025, M ≥ 4.5, literal mww: 15 and 53 rows. Common 0.25-M bins and annual counts preserve selection/evidence; they are not completeness-adjusted rates or relative hazard. Interactive view: Open this comparison.

4.2. Guided historical consultation

Simple view exposes map, statistics and printable-report routes while retaining the complete workspace (Figure 9). The same regional predicate and export engine serve both interfaces; Help explains coverage and terms.

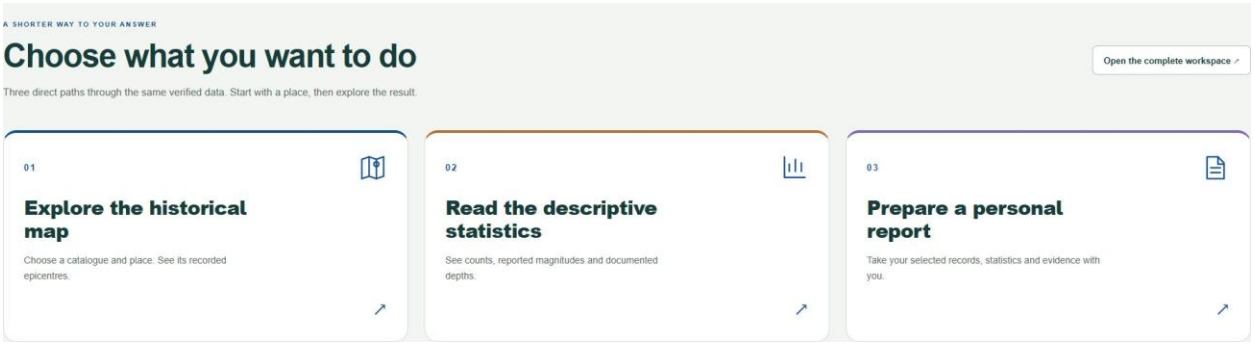


Figure 9. Three implemented Simple-view routes open historical maps, descriptive statistics and personalized reports through the complete workspace's filter/export engine. The full interface remains accessible. The capture demonstrates controls rather than measured usability gains. Interactive view: Open Simple view.

The default Top 10 retains four-source measurements (Figure 10A). In advanced single-source mode, SSN's Guerrero land class contains 21,167 M ≥ 3.5 rows in UTC 1900-2026 (partial 2026, unknown type). Its Top 10 starts with M 7.8 on 28 July 1957. Fourteen rows satisfy M ≥ 7; a 50-km radius yields ten direct pairs involving twelve (Figure 10B), without establishing independent repeats or hazard. A separate Python geometric oracle checks real/synthetic ranking and distance cases.



Figure 10. Complementary historical map extracts. (A) Default Top 10 ranks reported magnitude numbers across four catalogues, retaining source/type and both provenances for shared exact ComCat IDs; these are not homogeneous Mw measurements or reconciled physical events. Rank labels may be offset, with leader lines ending at the original epicentres. (B) SSN Guerrero land, UTC 1900-2026 (partial 2026, unknown magnitude type): fourteen rows satisfy M ≥ 7; ten pairs at ≤50 km involve twelve. Offshore context is disabled. Links/circles are selection geometry, not rupture recurrence or risk. Tables and CSV/JSON preserve criteria and hashes. Interactive views: A · Top10 · B · proximity.

4.3. Retained revisions and source roles

The pilot retains 236 public origin versions for 36 currently selected events (22 in 2017; 14 in 2022). All have UPDATE status; 205 belong to primary US lineages (Table 3). They are retained versions, not simultaneously active solutions. ISC-GEM authorship illustrates why distribution and origin authorship differ.

Table 3. Observed quantity	Value	Counting unit
Current-selected event identifiers	36	Captured events
Public origin-product versions	236	Retained versions
Versions in primary US origin lineages	205	Retained versions
Distributor source us	226	Public origin versions
Distributor source at / pt	6 / 4	Public origin versions
Origin author us / iscgem / missing	205 / 21 / 10	Public origin versions
Separate internal-origin / phase-data versions	15 / 203	Related products, excluded from primary analysis

Median endpoint displacement is 12.339 km, maximum 79.684 km; depth differences span -38.48 to 12.39 km. For us2000ahv0, retained endpoints change from 8.0 to 8.2 mww, 35.0 to 47.39 km depth and by 9.938 km epicentrally. These differences do not establish accuracy.

Thirteen of 31 same-type pairs change magnitude; five mixed-type pairs are excluded. Four cross 5.5 (Table 4), none the other tested thresholds. Among 30 mww pairs, earliest endpoints select 26 at $M \geq 5.5$ and latest select 30. Current-cohort admission at preferred $M \geq 5.5$ prevents interpreting upward crossings as an unbiased migration rate.

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ailq	5.4	5.5
us2000ap3b	5.4	5.6
us2000aqji	5.4	5.5
us1000arzf	5.4	5.5

Of 169 adjacent primary pairs, 117 change selected fields. Ten summary timestamps postdate every retained origin. For us2000ahv0, the primary branch ends in December 2017, another updates in May 2022, and the October-2026 summary concerns DYFI. These clocks do not form one origin-revision sequence.

4.4. Validation and observed coverage

Across 236 update boundaries, 472 cutoffs produce 4,720 queries agreeing with independent SQL; all 24 synthetic conditional projections agree. Against synthetic public truth, producer-time backcasting, filtering first and index-time substitution fail in 10, 11 and 10 overlapping cases. These are constructed failures, not operational rates. Stronger historical claims are declined by contract; empirical products contain no DELETE cases.

SSN records occupy 110 of 126 closed years (1900-2025), leaving 16 empty bins. Eight tied years yield 118 maximum-attaining rows. Partial 2026 includes 34,548 raw rows, 14,875 at $M \geq 3.5$ and 35 at $M \geq 5$. Reporting coverage establishes neither completeness nor rates/return periods.

5. Discussion

The system derives reusable information from heterogeneous selections while preserving multiplicity. Matching literal magnitude types does not ensure uniform measurement. Land filters omit offshore records; nearby context retains unassigned status. Spatial pairs, correlations and plate context establish neither rupture recurrence nor risk. Retained-version contracts support reproducible, weaker calculations when past public states are unknown. Existing history software remains prior art; no speed or accuracy superiority is claimed. Synthetic examples demonstrate ambiguity, and independent SQL validates the supplied evidence.

Static files and standard outputs facilitate migration, without guaranteeing twenty-year hosting or browser compatibility. Memory/external-service limits remain. Responsive checks are not usability evidence; independent users, broader cohorts and historical captures are needed. Future probabilistic models require justified completeness, magnitude scale, sampling and stationarity.

6. Conclusions

The observatory connects browser computation, documented inputs and traceable descriptive results for further study. Maps, comparisons, historical queries and exports retain source/type/geographic limits without a database server. Validation establishes reproducible calculations, not source accuracy, completeness, independence or past public states. It produces no earthquake forecast or validated risk estimate.

Code and data availability

The evaluated interface is release 2026.11.11.1. Distribution uses the study repository (https://github.com/gabrielhuav/Seismic_Data_Geosciences) and its public explorer (https://gabrielhuav.github.io/Seismic_Data_Geosciences/). Software, manifests and frozen evidence accompany

validation. Source hashes and capture receipts identify evaluated inputs and images; no archival DOI is asserted. All eighteen article targets restore the English selections shown in the figures.

The archive preserves 36 detail and four count/query responses; restore verified bytes rather than substitute live endpoints. SSN's full CSV SHA-256 is 0601c65544bc95ef865d3440e5e6d84249acdebd8edfb379669597096543f993. Its DOI identifies a mutable service, not this download. Credit belongs to SSN, Instituto de Geofísica, UNAM, without endorsement. Source-specific terms remain separate from MIT software, sql.js/SQLite notices, public-domain Natural Earth and ODC-BY PB2002 attribution.

Run python -m unittest discover -s tests and python -m scripts.validate_evidence; documented JavaScript/oracle checks validate browser calculations. Manifests distinguish observed inputs, geographic derivatives and synthetic controls.

CRediT authorship contribution statement

José Manuel Villa Vargas: Conceptualization; Formal analysis; Supervision; Writing - original draft; Writing - review & editing.

Gabriel Hurtado Avilés: Data curation; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI assistance

OpenAI Codex (OpenAI; GPT-6-based agent sessions, 6-11 November 2026) assisted literature discovery, code development, reproducible figure scripts, automated testing, manuscript drafting and document formatting. The explanatory diagram and data plots are rendered from documented inputs and maintained code. The interface panels derive from genuine integrated-browser captures of local previews, with capture-specific dates and hashes retained. Cropping and panel composition preserve the captured content; no generative image creation or retouching of observations is used.

References

Bird, P., 2003. An updated digital model of plate boundaries. Geochemistry, Geophysics, Geosystems 4(3), 1027. Article (https://doi.org/10.1029/2001GC000252).

European-Mediterranean Seismological Centre, n.d. Data contributors. Accessed 6 November 2026. Contributor directory (https://emsc-csem.org/Earthquake_data/contributors.php).

Gardezi, J., Bertossi, L., 2011. Query Answering under Matching Dependencies for Data Cleaning: Complexity and Algorithms. Author preprint, arXiv:1112.5908. Preprint (https://arxiv.org/abs/1112.5908).

Government of Canada, n.d. Open Government Licence-Canada, version 2.0. Accessed 6 November 2026. Licence (https://open.canada.ca/en/open-government-licence-canada).

Natural Earth, n.d.a. About Natural Earth. Geographic data and boundary policy. Accessed 6 November 2026. Project documentation (https://www.naturalearthdata.com/about/).

Natural Earth, n.d.b. Admin 1 - States, Provinces, 1:10 million, version 5.1.1. Accessed 8 November 2026. Official documentation (https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-1-states-provinces/).

Natural Resources Canada, n.d. Earthquakes in Canada. Open Government Portal, record 4cedd37e-0023-41fe-8eff-bea45385e469. Accessed 6 November 2026. Official dataset (https://open.canada.ca/data/en/dataset/4cedd37e-0023-41fe-8eff-bea45385e469).

Pintore, S., 2026. x-fdsnws-event: a reference implementation of a version-aware extension of the FDSNWS/event access model for reproducible seismic event catalogues. INGV software repository. Accessed 5 November 2026. Repository (https://github.com/INGV/x-fdsnws-event).

Sawires, R., Santoyo, M.A., Peláez, J.A., Corona Fernández, R.D., 2019. An updated and unified earthquake catalog from 1787 to 2018 for seismic hazard assessment studies in Mexico. Scientific Data 6, 241. Article (https://doi.org/10.1038/s41597-019-0234-z).

Servicio Sismológico Nacional, 2026a. Catálogo de sismos, México. Instituto de Geofísica, Universidad Nacional Autónoma de México. Downloaded 6 November 2026. Catalogue DOI (https://doi.org/10.21766/SSNMX/EC/MX).

Servicio Sismológico Nacional, 2026b. Aviso legal, uso de datos sísmicos y términos de libre uso. Observed 6 November 2026. Terms (http://www.ssn.unam.mx/aviso-legal/).

Sismanis, Y., Wang, L., Fuxman, A., Haas, P.J., Reinwald, B., 2009. Resolution-Aware Query Answering for Business Intelligence. Proceedings of the 25th IEEE International Conference on Data Engineering, 976-987. Publication (https://research.ibm.com/publications/resolution-aware-query-answering-for-business-intelligence).

sql.js contributors, n.d. sql.js, version 1.13.0: SQLite compiled to WebAssembly. Accessed 8 November 2026. Versioned source and documentation (https://github.com/sql-js/sql.js/tree/v1.13.0).

tectonicplates contributors, n.d. PB2002 plate and boundary GeoJSON conversion. Peter Bird model; conversion by Hugo Ahlenius/Nordpil and contributors. Commit 339b0c56563c118307b1f4542703047f5f698fae, ODC-BY 1.0. Pinned data and attribution (https://github.com/fraxen/tectonicplates/tree/339b0c56563c118307b1f4542703047f5f698fae); Licence (https://github.com/fraxen/tectonicplates/blob/339b0c56563c118307b1f4542703047f5f698fae/LICENSE.md).

Thompson, E.M., McBride, S.K., Hayes, G.P., Allstadt, K.E., Wald, L.A., Wald, D.J., Knudsen, K.L., Worden, C.B., Marano, K.D., Jibson, R.W., Grant, A.R.R., 2020. USGS near-real-time products; and their use; for the 2018 Anchorage earthquake. Seismological Research Letters 91(1), 94-113. Article (https://doi.org/10.1785/0220190207). Official publication record (https://www.usgs.gov/publications/usgs-near-real-time-products-and-their-use-2018-anchorage-earthquake).

U.S. Geological Survey, n.d.a. API Documentation; Earthquake Catalog. Accessed 5 November 2026. Documentation (https://earthquake.usgs.gov/fdsnws/event/1/).

U.S. Geological Survey, n.d.b. How Products Work in ComCat. Accessed 5 November 2026. Documentation (https://earthquake.usgs.gov/data/catalog/products.php).

U.S. Geological Survey, n.d.c. Product Distribution Layer; Indexer Processing. Accessed 5 November 2026. Documentation (https://usgs.github.io/pdl/indexer.html).

U.S. Geological Survey, n.d.d. libcomcat Application Program Interface. Accessed 5 November 2026. Documentation (https://github.com/usgs/libcomcat/blob/master/docs/api.md).

U.S. Geological Survey, n.d.e. ComCat Timeline. Historical source implementation and current developer-resource directory. Accessed 5 November 2026. Inspected implementation (https://github.com/jmfee-usgs/comcat-timeline/blob/master/src/Catalog.ts); Official directory (https://earthquake.usgs.gov/data/software/).

Villa Vargas, J.M., Hurtado Avilés, G., Climent Hernández, J.A., 2026. Cuando México tiembla: la historia contada por los datos. AZCATL Revista de Divulgación en Ciencias, Ingeniería e Innovación 4(6), 28-33. DOI: 10.24275/AZC2026E1004. Published antecedent (https://azcatl.azc.uam.mx/index.php/azcatl/article/view/75).

Weatherill, G.A., Pagani, M., Garcia, J., 2016. Exploring earthquake databases for the creation of magnitude-homogeneous catalogues: tools for application on a regional and global scale. Geophysical Journal International 206, 1652-1676. Article (https://doi.org/10.1093/gji/ggw232).

Wu, X., Poorthuis, A., Zurita-Milla, R., Kraak, M.-J., 2020. An interactive web-based geovisual analytics platform for co-clustering spatio-temporal data. Computers & Geosciences 137, 104420. Article (https://doi.org/10.1016/j.cageo.2020.104420).

Zúñiga, F.R., Reyes, M.A., Valdés, C., 2000. A general overview of the catalog of recent seismicity compiled by the Mexican Seismological Survey. Geofísica Internacional 39(2), 161-170. Public abstract and metadata consulted. Official record (https://repositorio.unam.mx/contenidos/4133493).